

ETL/ELT

TITTLE: Meaning and difference between ETL/ELT

Explanation:

ETL: (Extract transform load)

Process:

- ✓ Extract: data is collected from various sources(database APIs ,flat files)
- ✓ Transform: data is cleaned, formatted and transformed before loading into the target system
- ✓ Load: the transformed data is loaded into the target system usually a data warehouse

ETL:(Extract load transform)

Process:

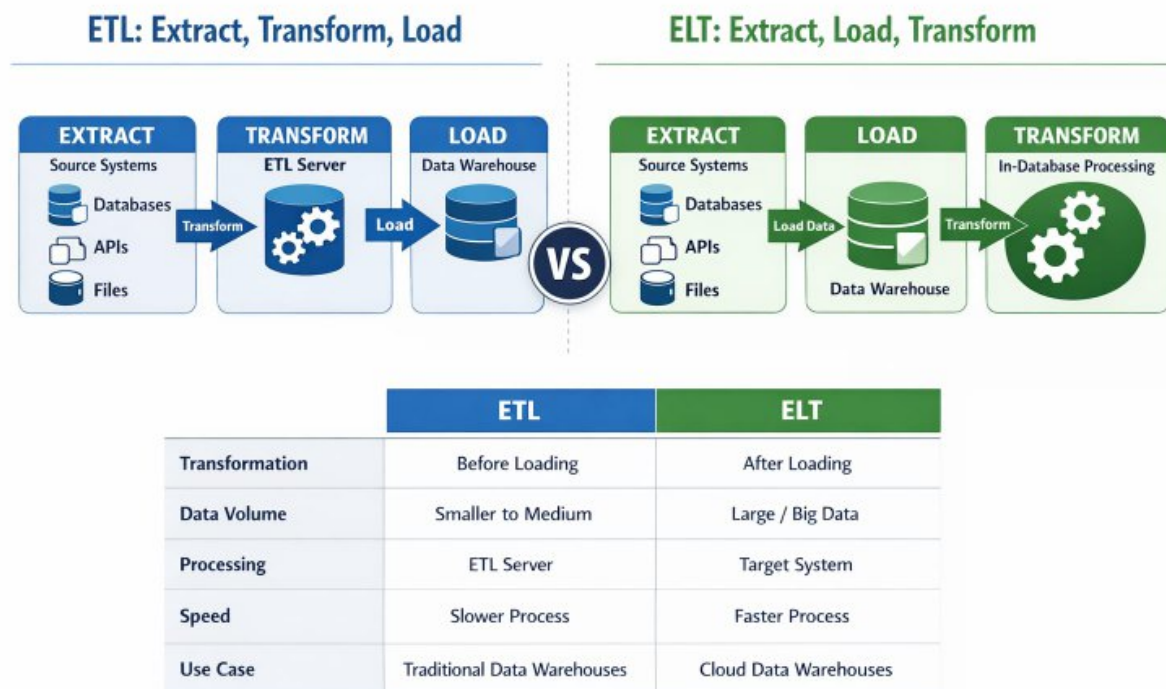
- ✓ Extract: data is collected from various sources
- ✓ Load: raw data is loaded into the target system (usually a cloud data warehouse or big data system)
- ✓ Transform: transformation happens inside the target system using its processing power

Differences between ELT/ETL:

ETL	ELT
Data transform before loading	Data transform after loading

Uses Separate ETL Server	Uses data warehouse power
Slower for larger data /Higher infrastructure cost	Faster for larger data /low cost

Diagram:



Interview questions:

- ✓ What is ELT/ETL?
 - o ELT— Extract load transform
 - o ETL---Extract transform load

- ✓ Main difference between ETL/ELT?
 - o Transformation timing
- ✓ Where does transformation happen in ETL/ELT?
 - o ETL— before loading
 - o ELT--- after loading
- ✓ Which is faster for big data and cloud data warehouse and semi structure data?
 - o ELT
- ✓ ETL/ELT tools
 - o ETL— informatical, talend, SSIS
 - o ELT----fivetran, stitch, matillion
- ✓ Explain ETL/ELT tools purpose
 - o Informatical— data clean and transform to stored warehouse
 - o Tlaned---data integration and data quality
 - o SSIS----data integration and data loading
 - o Fivetran----raw data automatically to stored cloud warehouse
 - o Stitch---simple and fast data replication
 - o Matillion---high performance data transformation